Definition of a Correction Ratio approach for Fission Matrix Temperature Interpolation and Monte Carlo Uncertainty Propagation Methods for Results Assessment

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Number of pages: 36
Number of tables: 4
Number of figures: 12

Abstract

The fission matrix (FM) method enables whole-core reactor transport calculations within seconds by interpolating a precalculated database of Monte Carlo (MC)–tallied coefficients across different operational parameters instead of running a direct MC calculation for every configuration of interest. This work focuses on the parametrization of the FM with respect to the temperature distribution within a micro core. Interpolation implicitly assumes that each coefficient depends only on the local temperature of cells where fission occurs. This assumption breaks down under large, spatially sharp temperature gradients. In this work, a Core Temperature Correction Ratio (CTCR) is introduced to compensate this effect.

The results require an uncertainty budget with two parts: the MC statistics of the reference case, and the uncertainty of the eigenvalue (k_{eff}) and eigenvector ($\nu\Sigma_f\phi$) of the *interpolated* matrix itself, a new object assembled from the database's finite-statistics matrices, which has no MC replicas of its own. We construct the latter by resampling the database and recomputing the interpolated (and corrected) result many times, comparing an uncorrelated baseline, an analytical model derived in this work, a measured-correlation model, and a brute-force reconstruction from 150 independent replicas. Validated against brute force across 12 test cases, the analytical model reproduces the brute-force k_{eff} uncertainty to within a mean absolute deviation of 4.6%, matching or exceeding the two alternative resampling models, while requiring no MC replicas beyond the nominal database already needed for interpolation. With both uncertainty sources combined, the CTCR improves the result in almost all cases: it removes most of the point-estimate k_{eff} discrepancy for sharply varying, highly localized temperature fields, whereas the improvement is smaller (and less necessary) for smoother temperature distributions. The local fission-source error follows the same pattern, improving substantially wherever a strong, spatially localized temperature gradient is present; where the correction succeeds, the residual discrepancy is generally consistent with the combined statistical noise of the reference case and the interpolated database.

Keywords — Fission matrix, correction ratio, uncertainty quantification, covariance, HPMR, temperature

I. INTRODUCTION

Monte Carlo (MC) methods provide the high-fidelity treatment of the neutron transport equation in reactor physics applications, at the cost of a statistical uncertainty that decreases only as the inverse square root of the simulated particle histories. Hybrid neutron transport methods based on the fission matrix (FM) formulation [1] address this cost by precalculating a database of FM coefficients under a set of reference system configurations and then interpolating and combining these coefficients to represent an arbitrary configuration within the database's validity range.

There are two different strategies for obtaining the FM of a single core configuration. The first one assembles the whole-core FM by running one fixed-source calculation for each "fundamental cell" of the system. As described in [2], the fundamental units are identified by exploiting the geometrical and material similarities between repeated units to limit the required database calculations. Reflectors and structural materials are deliberately excluded from the FM tally to preserve this similarity, and their influence is recovered afterward through dedicated boundary and discontinuity corrections. This is the approach of RAPID [2], a mature implementation yielding whole-core transport solutions within a few minutes at accuracy comparable to the underlying MC calculations, aided by dedicated techniques for FM homogenization [3], temperature interpolation [4], and correction of material and control-rod discontinuity effects [5, 6, 7]. The present work adopts the second strategy to obtain the whole-core FM for a single configuration, which consists in tallying the FM elements directly from a single whole-core criticality calculation in Serpent [8], so that reflectors, structural materials, and inter-assembly heterogeneity effects are already embedded in each coefficient, and a complete fission matrix is obtained from one calculation rather than assembled from fundamental-unit responses. The selection of the most suitable approach is case-specific: it depends on the reactor design and on the purpose of the analysis, keeping in mind the features of the two methods. The method used in RAPID, in fact, requires, where applicable, the recognition of the fundamental cells of the system, and the definition of multiple fixed-source MC calculations to build up one single whole-core FM. To guarantee the right convergence of the eigen-problem's results of the resulting FM, it is important that the mesh of the single fixed-source cases is fine enough to

capture all the local effects, at the same time a too small grid would be a burden memory-wise. This means that the dimension of the FM is not completely arbitrary but it must be a trade-off between the result-accuracy and the hard-disk storing capacity. Generally speaking the fixed-source calculations need less computational time and the uncertainty of the tallied FM elements is small if enough neutron histories are set. The whole-core FM deriving from a criticality calculation, instead, uses an already-converged neutron source, so that there is no direct dependence on the accuracy of the fundamental eigenvalue and eigenfunction. Because the coefficients are accumulated on the fly from a converged source, the within-cell spatial distribution of the fission source is represented much more accurately than with the fixed-source technique, so that much larger spatial cells can be used [2]. On the other hand, to obtain data with a low-enough uncertainty, especially for pin-wise cells, a high number of MC histories is necessary.

The second method is chosen because the goal is a simple workflow giving reliable FM results for an online core-monitoring tool for the Westinghouse eVinci reactor, where pin-wise accuracy is not needed.

The system studied here is a modified Heat-Pipe Micro-Reactor (HPMR) concept, and the operational variable of interest is the mesh cell temperature: the FM database is tallied at a set of uniform reference temperatures, and an arbitrary in-core temperature field is represented by interpolating each coefficient across that database. As a general approach widely adopted to deal with interpolations performed on temperature distributions, it is assumed that a coefficient a_{ij} depends only on the *local* temperature of the destination cell, i.e. the i-th cell. However, under a large, spatially sharp temperature gradient that assumption breaks down and the systematic error affecting the interpolated fission matrix can increase. To compensate it, we introduce a Core Temperature Correction Ratio (CTCR), calibrated once from two additional FMs, and show that it reduces the point-estimate discrepancy between interpolated and directly-simulated reference results across a range of test cases. The CTCR and this point-estimate comparison were first presented in a conference paper [9], which the present paper extends with an estimate of the statistical uncertainty of the interpolated results.

To better understand the systematic error and the improvement provided by the CTCR, a point-estimate comparison of the interpolated FMs and the reference one directly calculated with Serpent may not be enough to distinguish a real, uncor-

rected modeling bias, since both the database and the reference case are generated by Monte Carlo simulations, which always exhibit intrinsic statistical noise. Making that distinction requires quantifying the statistical uncertainty on both quantities being compared. This estimation is straightforward for the reference case: Serpent [8] reports its own statistical uncertainty for each tallied quantity, including both k_{eff} and the fission source, F. On the contrary, the results of the interpolated matrix are provided without any uncertainty: the eigenvalue and eigenvector of the matrix produced by interpolation (and CTCR correction) are new objects and, unlike a direct MC run, have no replicas of their own from which an uncertainty could be read off. The second aim of this work is to fill this gap with an inexpensive uncertainty estimate for the interpolated FM. This will allow us to better quantify any systematic error and the improvement provided by the correction ratio. The paper describes the database, the interpolation and the CTCR, then the two uncertainty sources and the four ways of estimating the database-side one, the error metrics, and finally the test cases and their results.

II. SYSTEM GEOMETRY DESCRIPTION

The system considered in this work is a modified version of the Heat-Pipe Micro-Reactor (HPMR) concept developed at Argonne National Laboratory under the NEAMS Micro-Reactor Application Drivers program [10]. It consists of 127 hexagonal, TRISO-fueled assemblies embedded in a graphite monolith. The configuration is a simplified model used to test the method. The control rods, the control drums and the beryllium reflector of the reference design are not represented: the assemblies are surrounded by a low-density mixture of helium and stainless steel up to a vacuum boundary. The temperature effects in the fueled region, which the CTCR addresses, are thus isolated from those of the components outside it. The core is treated as axially infinite, so that the analysis is carried out on a two-dimensional radial model throughout. The system is shown in Fig. 1.

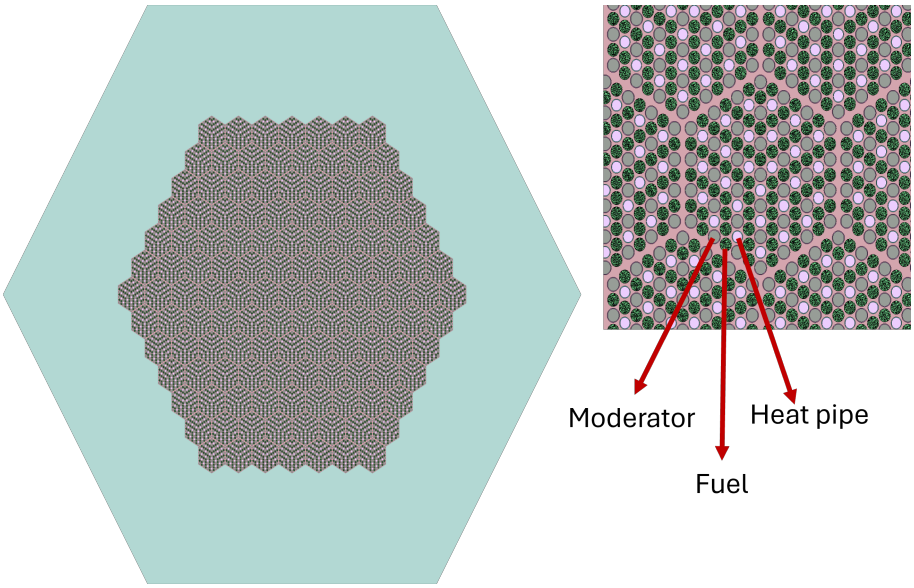

Fig. 1. System geometry.

III. METHODOLOGY

III.A. The Fission Matrix Method

The fission matrix (FM) method [1] recasts the k-eigenvalue transport problem into a compact linear-algebraic form by partitioning the fissile domain into N spatial regions and describing the transport process among them over a averaged single neutron generation. Applying the integral form of the k-eigenvalue transport equation over the volumes of a source region j and a destination region i yields an algebraic eigenvalue problem,

$$F_i = \frac{1}{k} \sum_{j=1}^{N} a_{ij} F_j, \qquad \Longleftrightarrow \qquad \mathbf{A}\,\mathbf{F} = k\,\mathbf{F}, \tag{1}$$

where F_j is the fission-neutron source integrated over region j, and the coefficient a_{ij} encodes the transport process yielding the expected number of fission neutrons born in region i per fission neutron emitted in region j. The $N \times N$ array of these coefficients is the fission matrix $\mathbf{A}$: its dominant eigenvalue is k_{eff} and the associated eigenvector is the region-wise fission-source distribution [1].

Since each coefficient is defined per emitted source neutron, the fission multi-

plicity is already embedded in it: a source neutron that induces a fission contributes $\bar{\nu}$ neutrons to the destination region. A key practical feature of the method is that the coefficients can be accumulated at negligible additional cost during an ordinary Monte Carlo criticality calculation, by tallying, over the active cycles, the region of birth of each fission neutron against the region of birth of its parent. In this work the full fission matrix (one region per fuel assembly, yielding a 127×127 operator) is obtained in this way from a single whole-core Serpent calculation. Hence, every coefficient reflects the complete core environment.

III.B. Database, Interpolation, and Correction Ratio

Rather than tallying a fission matrix directly for every temperature field of interest, the method precalculates a database of fission matrices at a set of uniform reference temperatures and interpolates within it. To keep the method practical, the database must stay compact without losing accuracy; the one used here contains the coefficients a_{ij} at only eight uniform temperatures spanning 300–1000 K in 100 K increments. For an arbitrary temperature field, each coefficient is first obtained by piecewise-linear interpolation between the two bracketing database temperatures:

$$a_{ij}(T_i) = a_{ij}^{-} + \frac{T_i - T_i^{-}}{T_i^{+} - T_i^{-}} \left(a_{ij}^{+} - a_{ij}^{-}\right), \tag{2}$$

where a_{ij}^{-} and a_{ij}^{+} are the values of a_{ij} at the database temperatures immediately below and above the arbitrary destination-cell temperature T_i.

Eq. (2) treats each coefficient as depending only on the local temperatures i, and therefore misses the influence of the surrounding cells. Under a large, spatially sharp gradient this is precisely where the interpolated matrix acquires a systematic error. The Core Temperature Correction Ratio (CTCR) defined in this work tries to compensate this by informing the numerical value of a generic matrix element a_{ij}, based on the pair of temperatures (T_i, T_j) through a correction ratio $\hat{r}_{ij}$. This correction is calibrated from two additional, directly simulated FMs, defined with "Dirac temperature distributions", shown in Fig. 2.

Because the correction must handle both a locally *hot* cell embedded in a colder surrounding and the reverse case, both Dirac FMs of Fig. 2 are used: one with the central assembly C driven to the hottest database temperature (1000 K) while the rest of the core sits at the coldest temperature (300 K), and one with the

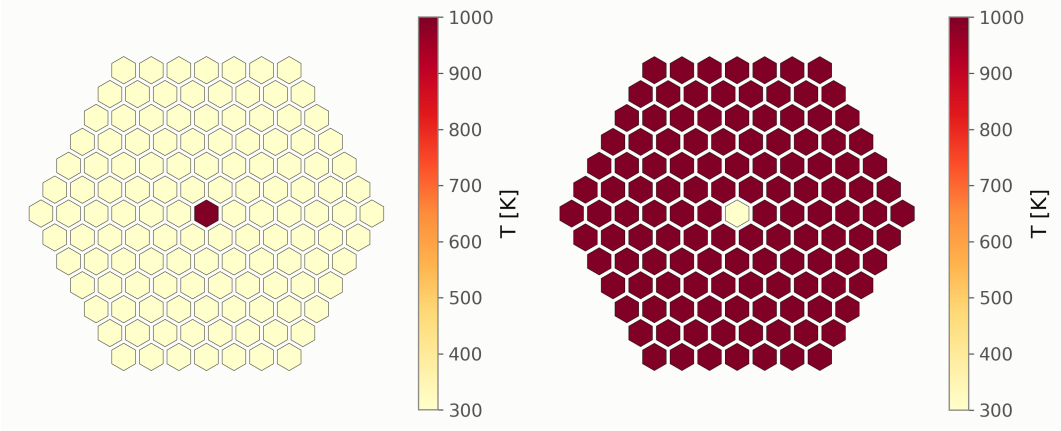

Fig. 2. The two additional cases with "Dirac temperature distributions", used to calibrate the CTCR.

assignment reversed. Comparing each Dirac FM against its own locally-interpolated counterpart gives two ratios, each a function of the geometric distance d between C and a generic cell j, including $j = C$ itself ($d = 0$) (exploiting the hexagonal core's sextant symmetry, so that only d, not direction, matters),

$$r^{+}(d) = \frac{a_{Cj}^{\mathrm{dirac},+}}{a_{Cj}^{\mathrm{interp}}}, \qquad r^{-}(d) = \frac{a_{Cj}^{\mathrm{dirac},-}}{a_{Cj}^{\mathrm{interp}}}, \tag{3}$$

where the superscript $+$ ($-$) denotes the hot-center (cold-center) Dirac case, and a_{Cj}^{interp} is the coefficient the ordinary interpolation of Eq. (2) would give for that same, uniform-elsewhere temperature field. Each ratio in isolation quantifies how much the local-temperature interpolation assumption misses, in the most extreme, the most spatially sharp case the database can represent; combined, the two make the correction able to handle a temperature difference of either sign.

For an arbitrary temperature field, the correction applied to a generic interpolated coefficient a_{kl}^{interp} is

$$\hat{r}_{kl}(\Delta T_{kl}, d) = \frac{\Delta T_{kl}}{\Delta T_{\mathrm{dirac}}}\left(r^{\pm}(d) - 1\right) + 1, \qquad \Delta T_{kl} = \begin{cases} T_k - T_l, & k \neq l, \\ T_k - \dfrac{1}{|\mathcal{N}(k)|}\displaystyle\sum_{n \in \mathcal{N}(k)} T_n, & k = l, \end{cases} \tag{4}$$

where $\Delta T_{\mathrm{dirac}} = 700$ K is the calibration temperature span common to both

Dirac cases, and $r^{+}(d)$ is used when $\Delta T_{kl} > 0$ (cell k hotter than l, or than its own first-ring neighborhood average $\mathcal{N}(k)$ when $k = l$) while $r^{-}(d)$ is used when $\Delta T_{kl} < 0$ — i.e., the calibration case matching the *sign* of the local temperature difference, evaluated at its own geometric distance d. For a diagonal coefficient ($k = l$, $d = 0$) this reuses $r^{\pm}(0)$ from the same Dirac calibration of Eq. (3), so no additional simulation or calibration data is needed to cover this case. Eq. (4) thus transfers and scales the calibration ratios $r^{\pm}(d)$ of Eq. (3) to any pair of cells and any temperature field. For $k \neq l$ the scaling is the actual temperature difference between the two cells divided by the calibration span. For $k = l$, source and destination coincide, and $r^{\pm}(0)$ measures how much the raw interpolation misses a_{kk} when the cell is surrounded by neighbors $\mathcal{N}(k)$ at a different average temperature $\frac{1}{|\mathcal{N}(k)|}\sum_{n\in\mathcal{N}(k)} T_n$; the scaling is then the difference between T_k and that average.

In this work the two variants compared are referred to as *raw* (interpolation only) and *corrected* (interpolation plus CTCR).

III.C. Two Sources of Uncertainty for k_{eff} and F

Comparing an interpolated (and CTCR-corrected) result (here generally indicated with X) to a direct reference calculation involves two statistically independent Monte Carlo noise sources, which combine in quadrature:

$$\sigma^2_{X,\text{tot}} = \sigma^2_{X,\text{ref}} + \sigma^2_{X,\text{interp}}. \tag{5}$$

(a) *Reference-case noise*, $\sigma_{X,\text{ref}}$: the statistical uncertainty Serpent reports directly for the single reference simulation being compared against.

(b) *Interpolated noise*, $\sigma_{X,\text{interp}}$: the noise that the result X inherits from the interpolated (and corrected) matrix, built from the 8 database calculations. This uncertainty source is not available as for (a): it has to be estimated. For the corrected variant a third, smaller term, the statistical uncertainty of the two calibration calculations, is evaluated separately in Sec. IV.D.

III.D. Quantification of $\sigma_{X,\text{interp}}$

In this work four different approaches are introduced to propagate the database uncertainty to the interpolated FM results. Three of these methods share the same

procedure: the eight-temperature database is perturbed with random noise, the interpolator and correction of Eq. (4) are reapplied, and the procedure is repeated many times. The spread of the resulting $k_{\text{eff}}^{\text{interp}}$ and fission source F^{interp} over these repetitions is the database-side uncertainty.

Methods (i)–(iii) below differ in how the random perturbation is enforced. The perturbation acts on the fission-matrix coefficients a_{ij} of each database matrix, before interpolation and correction: each realization replaces a_{ij} with $a_{ij} + \delta_{ij}$. Indicating the perturbations of the retained coefficients of one database matrix (defined below) as a vector $\boldsymbol{\delta}$ drawn from a zero-mean multivariate normal distribution, $\boldsymbol{\delta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, the covariance matrix $\boldsymbol{\Sigma}$ factors as

$$\boldsymbol{\Sigma} = \mathbf{D}\,\boldsymbol{\rho}\,\mathbf{D}, \tag{6}$$

where $\mathbf{D}$ is the diagonal matrix of the per-coefficient standard deviations reported by Serpent, and $\boldsymbol{\rho}$ is the correlation matrix. The amplitude $\mathbf{D}$ is common to methods (i)–(iii) — it is read directly from the Serpent-reported uncertainties of the database — so they differ only in what they assume for $\boldsymbol{\rho}$.

$\boldsymbol{\Sigma}$ is defined separately at each database temperature. It is a $p \times p$ matrix, where $p = 6539$ is the number of retained coefficients out of $N^2 = 16\,129$: coefficients smaller than 10^{-4} times the largest element of the 300 K matrix are discarded, and the same p positions are used at all temperatures (the discarded coefficients carry less than 0.04% of the matrix sum and keep their nominal values). No coupling between temperatures is modeled (see below), so the joint covariance of the eight matrices is the $8p \times 8p$ block-diagonal matrix with the eight $\boldsymbol{\Sigma}(T)$ along its diagonal, which never needs to be assembled.

Operationally, for methods (i) and (ii) the perturbation is drawn column by column. Because the covariance structure derived in Sec. III.E couples only coefficients sharing a source column, $\boldsymbol{\Sigma}$ is block diagonal, with one $p_j \times p_j$ block per source region j, where p_j (between 41 and 67) is the number of retained coefficients in column j. Factorizing the dimensionless correlation block as $\boldsymbol{\rho}_j = \mathbf{C}_j \mathbf{C}_j^{\top}$ by Cholesky decomposition, the m-th realization of the perturbation of column j at database temperature T is

$$\boldsymbol{\delta}_{:,j}^{(m)}(T) = \mathbf{D}_j(T)\,\mathbf{C}_j(T)\,\mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{p_j}), \tag{7}$$

so that $\mathrm{Cov}(\boldsymbol{\delta}_{:,j}) = \mathbf{D}_j \boldsymbol{\rho}_j \mathbf{D}_j$ by construction. A single vector $\mathbf{z}$ is shared by all coefficients of the same column, which is what imposes the correlation; the marginal standard deviation of each individual coefficient remains σ_{ij} regardless of $\boldsymbol{\rho}$. The factorizations are performed once and reused across all realizations. For method (iii), whose measured correlation matrix also couples different columns, the full $p \times p$ matrix is factorized and the perturbation of all coefficients is drawn at once. The eight database matrices come from separate Monte Carlo calculations, in which the fission matrix is tallied directly in continuous energy, and no cross-section library generated by Monte Carlo is shared between them. Their noise is expected to be uncorrelated, and the brute-force replicas bear this out: over the 150 replicas available at each temperature, the correlation coefficient between the values of the same coefficient at two adjacent temperatures, averaged over all retained coefficients, lies between -0.0007 and $+0.0013$, consistent with zero. The perturbations of the eight matrices are therefore drawn independently, and no inter-temperature covariance is modeled.

Each perturbed database is then passed through the full pipeline: interpolation of Eq. (2), correction of Eq. (4), and solution of the eigenproblem of Eq. (1), yielding one realization of $k_{\mathrm{eff}}^{\mathrm{interp}}$ and of the normalized fission source. The procedure is repeated $M = 500$ times, and σ_{interp} is taken as the standard deviation of the resulting ensemble. Propagating the database covariance through a linearized sensitivity expression would be cheaper, but k_{eff} and the fission source are nonlinear functions of the database coefficients, through the eigenvalue problem; resampling the full pipeline avoids any linearization of this dependence. The correction ratios $r^{\pm}$ of Eq. (3) are computed once and kept fixed in all realizations. Their statistical uncertainty, which comes from the two calibration calculations, is therefore not part of σ_{interp}; it is evaluated separately in Sec. IV.D.

The four approaches are listed below. Methods (i)–(iii) follow the resampling procedure just described and differ only in the correlation matrix $\boldsymbol{\rho}$ of Eq. (6): (i) neglects the correlation, (ii) predicts it in closed form (Sec. III.E), and (iii) measures it from a single run. Method (iv) does not use Eq. (6): it rebuilds the whole database from independent replicas, and is the reference against which (i)–(iii) are validated in Sec. IV.B.

(i) **Uncorrelated resampling ($\boldsymbol{\rho} = \mathbf{I}$).** Each retained coefficient is perturbed

independently using only its own MC-reported variance, $a_{ij} \to a_{ij} + \delta_{ij}$ with $\delta_{ij} \sim \mathcal{N}(0, \sigma_{ij}^2)$. This is equivalent to setting $\boldsymbol{\rho}$ to the identity matrix and is one of the two assumptions considered in prior FM uncertainty-quantification work [11]. It is the baseline against which the effect of modeling the correlation is measured.

(ii) **Analytical model ($\boldsymbol{\rho} = \boldsymbol{\rho}_{\mathbf{pred}}$).** The correlation matrix is predicted in closed form, $\boldsymbol{\rho}_{\text{pred}}$, derived directly from the branching structure of the fission process. To the best of our knowledge, this closed-form prediction of the fission-matrix correlation structure from the branching process, and its use to build the interpolated matrix uncertainty, has not been reported before: prior work [11] considered two limiting cases, no correlation between coefficients and a full positive correlation within each column, whereas the derivation of Sec. III.E predicts that coefficients sharing a column are negatively correlated. The approach requires no Monte Carlo replicas beyond the single nominal database already needed for interpolation.

(iii) **Correlated resampling ($\boldsymbol{\rho} = \hat{\boldsymbol{\rho}}$).** The correlation matrix is instead obtained from the cycle-by-cycle tallies of a single MC run, treating each active cycle as a pseudo-replica of the same underlying random vector. Since Serpent does not provide the fission matrix on a per-cycle basis, this required a modification of the source code to tally and print each active cycle's fission matrix separately. Because the run has 500 active cycles, fewer than the $p = 6539$ coefficients, the sample correlation matrix is singular; it is regularized with Ledoit–Wolf shrinkage [12].

This $\hat{\boldsymbol{\rho}}$ is measured once, at 300 K, and reused unchanged at every other database temperature. This choice is tested in Sec. III.E.3.

(iv) **Brute force.** The full eight-temperature database is rebuilt from scratch $N = 150$ times, with one independent MC replica per temperature and per repetition, reapplying the interpolator and the same correction each time. This bypasses the $\boldsymbol{\Sigma} = \mathbf{D}\boldsymbol{\rho}\mathbf{D}$ construction entirely: all correlations are present in the replicas by construction, with no modeling assumption. It is the most direct and most expensive estimate.

III.E. Deriving the Analytical Correlation Model

Coefficients tallied within a single Monte Carlo criticality calculation are not, in general, statistically independent, because the number of neutron histories simulated per active cycle is fixed, a fact responsible for well-documented biases and correlations in cycle-based Monte Carlo estimators [13]. For two FM coefficients sharing the same source column j (i.e., a_{ij} and a_{lj}, $i \neq l$), this mechanism can be made explicit and turned into a closed-form prediction.

The derivation rests on how histories are tracked in the present calculations. Capture and fission are analog, no splitting or weight windows are used, and a history is terminated when it causes fission. The only non-analog element is Serpent's default implicit treatment of (n, xn) reactions, which changes the weight of a neutron but not the number of histories; its effect is quantified after Eq. (13). Two properties follow: a history contributes to at most one destination cell, and different histories of the same cycle are independent of one another given the source distribution, since simulated particles do not interact. Under implicit capture or implicit fission, where a history survives an absorption or a fission and may fission again, or under particle splitting, where the descendants of one source neutron are not independent, these two properties would have to be revisited.

At the start of a cycle, the number of source neutrons born in column j, N_j, is fixed and inherited from the fission bank of the previous cycle. Each of these N_j histories ends in exactly one of $N+1$ mutually exclusive outcomes: a fission in one of the N cells, or no fission at all (capture or leakage). If π_i is the probability that a history causes fission in cell i, the probability that it ends in any other way is $1-\pi_i$. The outcome of a single history therefore follows a categorical distribution, and the numbers of histories ending in each outcome follow a multinomial distribution.

If a history causes fission, all $\bar{\nu}$ secondaries are born together at that one destination. Writing the contribution of history s to destination i as $X_s^{(i)} = \bar{\nu}\, I_s(i)$, where $I_s(i)$ is 1 if history s fissions in i and 0 otherwise (a Bernoulli variable with mean π_i), the coefficient a_{ij} is the per-source-neutron average:

$$a_{ij} = \frac{1}{N_j} \sum_{s=1}^{N_j} X_s^{(i)}. \tag{8}$$

Since a history fissions at most once, $I_s(i) \cdot I_s(l) = 0$ for $i \neq l$. Using the

definitions $\mathrm{Var}(U) = E[U^2] - E[U]^2$ and $\mathrm{Cov}(U,V) = E[UV] - E[U]E[V]$, where $E[\cdot]$ is the expected value, and $E[I_s(i)^2] = E[I_s(i)] = \pi_i$ (an indicator equals its own square), one obtains for a single history,

$$\mathrm{Var}\left(X_s^{(i)}\right) = \bar{\nu}^2\,\pi_i(1-\pi_i), \quad \mathrm{Cov}\left(X_s^{(i)}, X_s^{(l)}\right) = \bar{\nu}^2\left[\underbrace{E\left[I_s(i)I_s(l)\right]}_{=0} - \pi_i\pi_l\right] = -\bar{\nu}^2\,\pi_i\pi_l, \tag{9}$$

and summing N_j independent, identically distributed histories (thus summing over s) and dividing by N_j as in Eq. (8),

$$\mathrm{Var}(a_{ij}) = \frac{\bar{\nu}^2\,\pi_i(1-\pi_i)}{N_j}, \qquad \mathrm{Cov}(a_{ij}, a_{lj}) = -\frac{\bar{\nu}^2\,\pi_i\pi_l}{N_j}, \qquad i \neq l. \tag{10}$$

The covariance is *negative* for every pair of destinations sharing a source column, and this follows directly because a fixed number of histories is split, exhaustively and mutually exclusively, among the possible outcomes, so that more of N_j going to i automatically means less going to l. Dividing the covariance by the geometric mean of the two variances, the common factor $\bar{\nu}^2/N_j$ cancels exactly, leaving a correlation that depends only on the destination probabilities themselves, not on N_j, which is a Serpent run-time setting, not a physical property of the fission process:

$$\rho_{\mathrm{pred}}(i,l) = \frac{\mathrm{Cov}(a_{ij}, a_{lj})}{\sqrt{\mathrm{Var}(a_{ij})\,\mathrm{Var}(a_{lj})}} = -\sqrt{\frac{\pi_i}{1-\pi_i}\cdot\frac{\pi_l}{1-\pi_l}}. \tag{11}$$

This is the Pearson correlation coefficient: $\rho \in [-1,1]$ measures the strength and the direction of the linear relationship between two variables, and a negative value means that a positive fluctuation of one coefficient tends to be accompanied by a negative fluctuation of the other. Since $\pi_i \ll 1$, the predicted correlations are weak: at 300 K their mean over all pairs sharing a column is -0.0031, and the strongest is -0.08. Equation (11) is written in terms of the destination probabilities π_i, which are not directly tallied; what Serpent reports is the mean coefficient $a_i = \bar{\nu}\pi_i$ (the expected number of secondaries reaching i per source neutron, i.e., $\pi_i = -\kappa a_i$ with $\kappa = -1/\bar{\nu}$). Substituting $\pi_i = -\kappa a_i$ into Eq. (11) and simplifying gives the closed

form used for method (ii),

$$\rho_{\mathrm{pred}}(i,l) = \kappa \, \frac{\sqrt{a_i a_l}}{\sqrt{(1+\kappa a_i)(1+\kappa a_l)}}, \qquad \kappa = -\frac{1}{\bar{\nu}}, \tag{12}$$

which requires only $\bar{\nu}$ (from `NUBAR` in `_res.m`) and the mean values of the nominal database matrix itself, with no additional MC replicas. (The multinomial variance and covariance results used in Eqs. (9-10) are standard; see, e.g., Ref. [14].)

Two simplifications in this derivation deserve a comment. The multiplicity $\bar{\nu}$ has been treated as a single constant, whereas ν depends on the incident neutron energy and each cell sees a slightly different spectrum. Each history has also been given the same weight, whereas Monte Carlo weights are not exactly unity.

Neither affects the amplitude of the perturbation. The matrix $\mathbf{D}$ of Eq. (6) is read from the statistical errors Serpent reports for each coefficient, and those already contain the local spectrum and the weights of the actual simulation. The multiplicity enters only through the conversion $\pi_i = a_i/\bar{\nu}$ needed to evaluate Eq. (12), because the common factor $\bar{\nu}^2/N_j$ cancels in Eq. (11). The multiplicity entering the tally is also an expected value: Serpent increments the fission-matrix bin by the neutron weight times $\bar{\nu}(E_0)$, evaluated at the incident energy of the neutron inducing that fission, and the integer number of secondaries sent to the fission bank never enters the score. The form $X_s^{(i)} = \bar{\nu}\, I_s(i)$ therefore describes the estimator itself, and the variance of ν around its mean does not contribute to $\mathrm{Var}(a_{ij})$.

Both simplifications enter the derivation at the same place. Neither changes the covariance of Eq. (10), which vanishes term by term as long as a history fissions once at most, whatever weight and multiplicity it carries; both change only the variance that normalizes it. Writing c_i^2 for the relative variance of the per-event score $w\,\bar{\nu}(E)$, conditional on a fission in i, Eq. (11) becomes

$$\rho_{\mathrm{pred}}(i,l) = -\sqrt{\frac{\pi_i}{1-\pi_i+c_i^2} \cdot \frac{\pi_l}{1-\pi_l+c_l^2}}\,, \tag{13}$$

and the model used here is the case $c_i^2 = 0$. For the present calculations c^2 is small. The energy dependence of $\bar{\nu}$ contributes 1.5×10^{-4}, obtained from the eleven-group $\bar{\nu}$, fission cross section and flux of the reference calculation, and corresponding to a 1.24% RMS spread of $\bar{\nu}$ across the fission spectrum. The weights contribute a com-

parable amount, since they depart from unity only through the implicit treatment of (n, xn) reactions, which here accounts for 1.2×10^{-4} of the total source weight. Propagating $c^2 = 1.5 \times 10^{-4}$ through the full pipeline changes σ_{interp} by 0.004% on k_{eff} and by 0.002% on the fission-source RMS. Assigning conservatively the whole 1.24% spread as an error on the $\bar{\nu}$ of Eq. (12) changes them by 0.33% and 0.14%; for reference, the core-average $\bar{\nu}$ moves by 0.018% over the whole 300–1000 K database. The aim of Eq. (12) is to provide a justified definition of the correlation rather than an exact model of it, and these effects are well below the accuracy it reaches against the brute-force reconstruction in Sec. IV.B.

III.E.1. Uncorrelated Treatment of Coefficients in Different Columns

The derivation above does not apply to coefficients that do *not* share a column, a_{ij} and $a_{kj'}$ with $j \neq j'$. It would be convenient, but not quite correct, to argue that N_j and $N_{j'}$ are simply independent: the total population $N = \sum_j N_j$ is fixed by the simulation's particle count per cycle, so source regions compete for a shared budget and $\mathrm{Cov}(N_j, N_{j'}) \neq 0$ in general. This coupling, however, need not propagate to the *normalized* coefficients themselves. By the law of total covariance,

$$\mathrm{Cov}(a_{ij}, a_{kj'}) = \underbrace{E\big[\,\mathrm{Cov}(a_{ij}, a_{kj'} \mid N_j, N_{j'})\big]}_{\text{(I)}} + \underbrace{\mathrm{Cov}\left(E[a_{ij} \mid N_j],\, E[a_{kj'} \mid N_{j'}]\right)}_{\text{(II)}}. \quad (14)$$

where term (I) vanishes because histories born in the disjoint columns j and j' are conditionally independent given $N_j, N_{j'}$ (again, no history-history interaction is assumed). Term (II) vanishes because $E[a_{ij} \mid N_j] = \pi_i$ is, by construction of the normalized estimator of Eq. (8), a constant that does not depend on the realized value of N_j, so its covariance with anything is zero. This rules out one specific channel, competition for a shared per-cycle particle budget, as a source of cross-column correlation; it is not a claim that cross-column correlation is exactly zero for every other conceivable reason (e.g., spatially correlated local flux fluctuations, or the cycle-to-cycle propagation of the fission source itself [15]).

III.E.2. Check on Independent Replicas

The correlation model is checked directly on the 150 independent replicas of the brute-force set, at each database temperature. The predicted correlations are too weak to be resolved pair by pair, since with 150 replicas a single measured correlation coefficient has a statistical uncertainty of about 0.08, but their effect accumulates in a sum over a column. The standard deviation of each column sum, $\sum_i a_{ij}$ over the retained coefficients, was therefore measured over the replicas and compared with the value predicted from the standard deviations of the individual coefficients, without correlation and with Eq. (12), as in replica-based checks of Monte Carlo variance [16, 17]. The result is shown in Fig. 3. Without correlation the prediction is too large in every column and at every temperature, by about 30% on average. With Eq. (12) the points lie along the identity line, and the average over the columns differs from the measurement by about 2% at most.

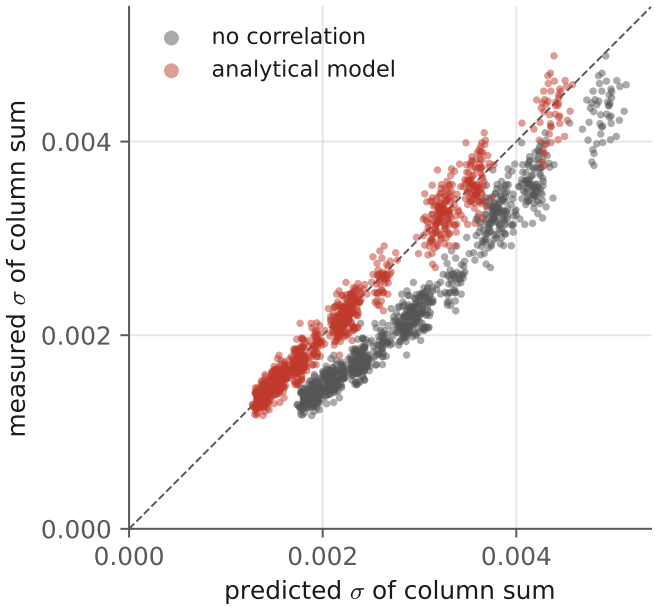

Fig. 3. Standard deviation of the column sums of the retained coefficients: value measured over the 150 independent replicas against the value predicted from the standard deviations of the individual coefficients, with no correlation (gray) and with the correlation of Eq. (12) (red). One point per source column and database temperature; on the dashed line the prediction equals the measurement.

The same replicas show no net correlation between different columns: the variance of the sum of all retained coefficients is 0.95 ± 0.05 times the sum of the column variances. Pairs of coefficients that share a destination row do show a small pos-

itive mean correlation, about one tenth of the same-column value, which method (ii) neglects. The effect of this approximation on the propagated uncertainties is assessed in Sec. IV.B, where the three methods are compared with the brute-force reconstruction.

These results also bear on fission matrices assembled from fixed-source calculations, as in RAPID. The argument above applies to any set of analog histories started from one source region, but the resulting correlation also depends on how the coefficients are tallied, normalized and reused for repeated units: for their fixed-source database, He and Walters [11] found a full positive correlation within each column to be more accurate than no correlation. The sign of the within-column correlation is therefore not universal, and the correlation model has to be chosen for the tallying scheme at hand.

III.E.3. Temperature Invariance of the Measured Correlation Matrix

Method (iii) assumes that the correlation matrix measured at 300 K also applies at the other database temperatures. The simplest test of this assumption is to measure $\boldsymbol{\rho}$ independently at the opposite end of the database range, 1000 K, and ask whether the two matrices differ by more than the statistical noise inherent in estimating either one.

Differences between correlation matrices are quantified here through the Frobenius norm, which treats a $p \times p$ matrix as a single vector of its entries and measures its Euclidean length,

$$\|\mathbf{M}\|_F = \sqrt{\sum_{a=1}^{p}\sum_{b=1}^{p} M_{ab}^2}\,. \tag{15}$$

The distance between two matrices is the norm of their difference; normalizing by the norm of the reference matrix $\mathbf{A}$ gives the dimensionless *relative* Frobenius distance used throughout,

$$d_F(\mathbf{A},\mathbf{B}) = \frac{\|\mathbf{A}-\mathbf{B}\|_F}{\|\mathbf{A}\|_F}\,, \tag{16}$$

so that $d_F = 0$ for identical matrices and $d_F \sim 1$ for matrices differing by an amount comparable to their own magnitude.

For this test each matrix is measured from the active cycles of 50 independent runs at that temperature (25 000 cycles in all), without shrinkage. Table I reports

this distance between the 300 K and 1000 K matrices, alongside an internal noise floor for each temperature: the distance between two estimates of the *same* matrix, obtained by splitting that temperature's own cycle data into two halves and measuring $\boldsymbol{\rho}$ from each. The noise floor is the distance one would obtain even if the true correlation matrix was exactly temperature-invariant, arising purely from the finite statistics of the estimate.

TABLE I
Simple Check of Temperature Invariance for the Measured Correlation Matrix

Comparison	Relative Frobenius distance
$\boldsymbol{\rho}$(300 K) vs. $\boldsymbol{\rho}$(1000 K)	0.643
$\boldsymbol{\rho}$(300 K), split-half internal noise floor	0.909
$\boldsymbol{\rho}$(1000 K), split-half internal noise floor	0.909
Split-half floor $\times 1/\sqrt{2}$ (full sample)	0.643

The noise floor is computed from estimates that use half of the data, whose statistical noise is larger by a factor $\sqrt{2}$ than that of the full-sample estimates compared in the first row. Scaled to the full sample, to first order in the noise, it becomes $0.909/\sqrt{2} = 0.643$, equal to the distance between the 300 K and 1000 K matrices: the two differ by the amount expected from statistical noise alone, and no temperature dependence is detectable at the resolution of the estimate. The resolution is limited, since distances of 0.6–0.9 mean that the measured matrices are dominated by noise; the test excludes only large changes of the correlation structure with temperature. The analytical model (ii) does not rely on this assumption: $\boldsymbol{\rho}_{\mathrm{pred}}$ is rebuilt at each database temperature from that temperature's own matrix and $\bar{\nu}$.

The same distance can be evaluated for the two Dirac configurations from which the correction ratios r^+ and r^- are computed, in which the temperature is not uniform. The cycle-by-cycle matrices needed for $\hat{\boldsymbol{\rho}}$ are not available for these runs, so the comparison is made with the analytical model, which can be evaluated on any matrix. The predicted correlation matrix of each configuration differs from that of the uniform core at the temperature of the surrounding assemblies by $d_F = 0.0013$–0.0014, the same value obtained by changing only the random seed (Table II). A single perturbed assembly therefore does not change the predicted correlation structure.

III.F. Definition of Error Metrics

Two plain metrics are used throughout, deliberately without a covariance-weighted quadratic form such as a Mahalanobis distance: they are less sensitive to the *spatial pattern* of a residual, but directly interpretable, which is what matters for our case-by-case comparison. For k_{eff},

$$\Delta k_{\text{eff}} = k_{\text{eff}}^{\text{interp}} - k_{\text{eff}}^{\text{ref}}, \qquad z = \frac{\Delta k_{\text{eff}}}{\sigma_{\text{tot}}}. \tag{17}$$

The fission source F is normalized to unit sum over the 127 cells giving $\tilde{F}_i$. Its local relative deviation in each zone is defined as

$$\delta\tilde{F}_i = \frac{\tilde{F}_i^{\,\text{interp}} - \tilde{F}_i^{\,\text{ref}}}{\tilde{F}_i^{\,\text{ref}}} \times 100\%, \tag{18}$$

and the overall discrepancy between the interpolated and reference source distributions is summarized by the root-mean-square (RMS) of the zone-wise relative deviations,

$$\text{RMS} = \sqrt{\left\langle \delta\tilde{F}_i^{\,2} \right\rangle_i}. \tag{19}$$

Every acceptance criterion in Sec. IV is $|z| < 2$ with the σ_{tot} of Eq. (5); where a figure shows the two terms separately, the reference term is drawn as a $\pm 2\sigma_{\text{ref}}$ band and the interpolated term as the cloud of realizations. σ_{interp} in Eq. (5) uses the analytical model (ii) throughout Sec. IV, computationally validated next.

IV. RESULTS

The pipeline is applied to twelve validation cases, each with a different in-core temperature field imposed on the HPMR model, reported in Fig. 4 and Fig. 5. They are chosen to span a wide range of spatial sharpness, since it is precisely under sharp gradients that the local-temperature interpolation assumption – and hence the CTCR – is most severely tested. The set comprises two spatially sharp "Dirac-like" perturbations, in which a single central assembly is offset in temperature from an otherwise uniform core; a six-assembly and two ring-shaped hot-spot patterns; a smooth linear gradient across the core; and six power-shaped profiles of varying

peak sharpness and radial offset. The two Dirac cases are the configurations used to calibrate the CTCR (Fig. 2) and are evaluated with the same calculations, so they test the correction on its own calibration data. The other ten cases were not used to compute the correction ratios $r^{\pm}$ and test their transfer to other assembly pairs and temperature levels: in the six-assembly and ring cases the heated assemblies lie in the second and fourth rings around the center, in two of them together with the central assembly, and the power-shaped profiles have minimum temperatures between about 350 and 670 K. The set also includes a single assembly at 1000 K in the fourth ring on a background of 350–450 K (power spike), so the set covers off-center perturbations and backgrounds other than 300 K. Each case is evaluated in both the *raw* (interpolation only) and *corrected* (interpolation plus CTCR) variants against a direct Serpent reference calculation of the same temperature field.

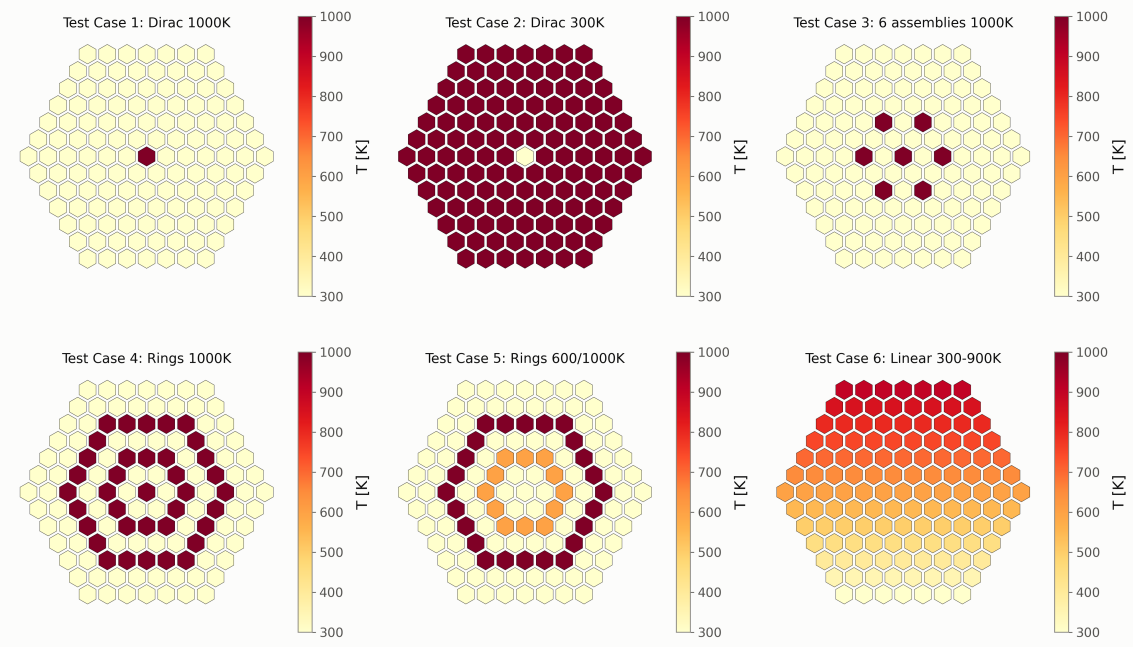

Fig. 4. Temperature distributions of the reference cases (part 1).

IV.A. Reference-Case Uncertainty

Figure 6 shows the reference case's uncertainty σ_{ref} (see Eq. (5)). Averaged over the twelve cases, it is 9.7 pcm on k_{eff} and 0.309% (RMS) on the fission source, and is essentially uniform across cases; the one mild outlier is the power shape

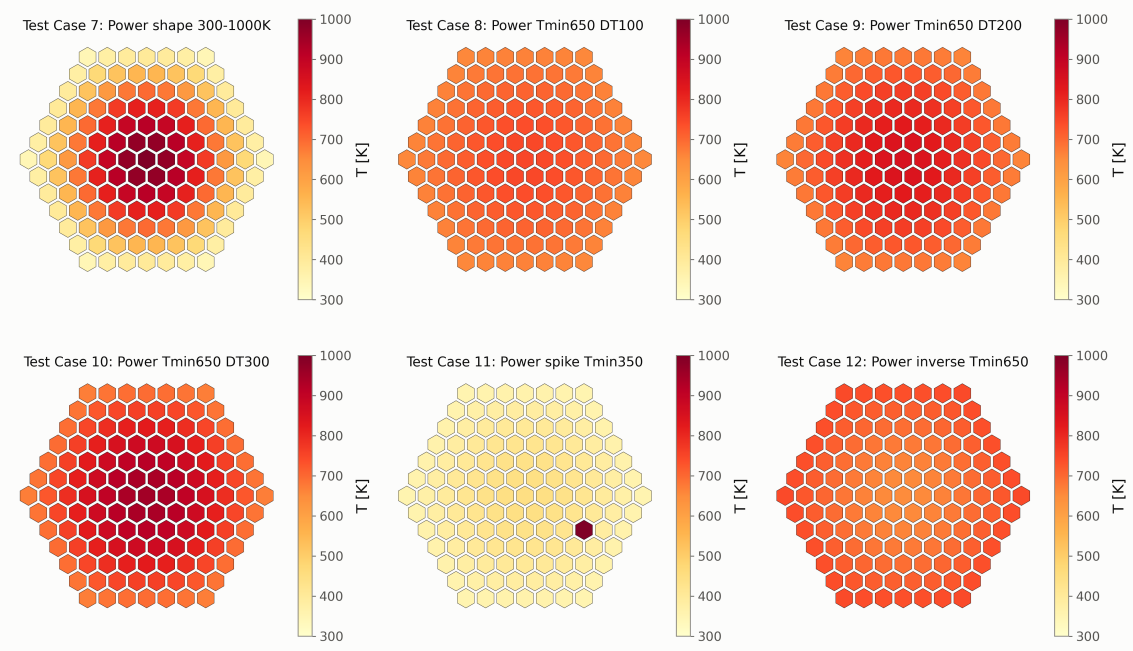

Fig. 5. Temperature distributions of the reference cases (part 2).

300–1000 K case (12.5 pcm, 0.42%), whose broader temperature profile produces a less peaked fission-source distribution and hence a statistically noisier detector tally. This is only half of the noise budget of Eq. (5): on its own it says nothing about the uncertainty introduced by the interpolation, quantified next.

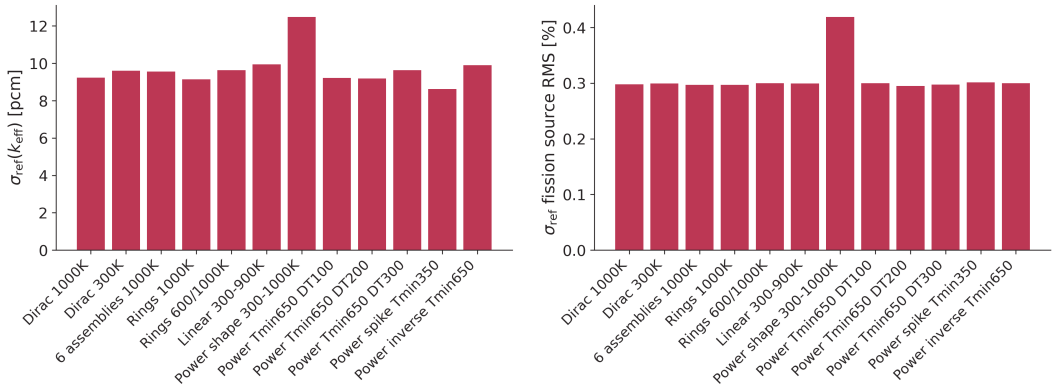

Fig. 6. Reference-case Monte Carlo uncertainty σ_{ref}: k_{eff} (left) and fission-source RMS (right), for all twelve cases.

IV.B. Validating the Database-Noise Models Against Brute Force

Figure 7 compares the three inexpensive approaches: uncorrelated (i), analytical (ii), and measured-correlation (iii), against the brute-force reconstruction (iv), for the corrected variant of all twelve cases, on both observables. Agreement is summarized by the mean absolute deviation from brute force across the twelve cases. The ordering of the interpolated uncertainties σ_{interp} predicted by the three models, excluding the brute-force results, which are used here as reference, is consistent with expectations in almost all the test cases. The uncorrelated model systematically yields the largest σ_{interp}, and therefore the widest confidence intervals, the only exception being k_{eff} in Test Case 4; the analytical covariance model gives the smallest values, while the covariance matrix obtained by considering every cycle of the MC simulation, lies in between. This single exception can be ascribed to statistical noise: the database is resampled randomly from a Gaussian distribution centered on the nominal values, so fluctuations of this magnitude are not unexpected.

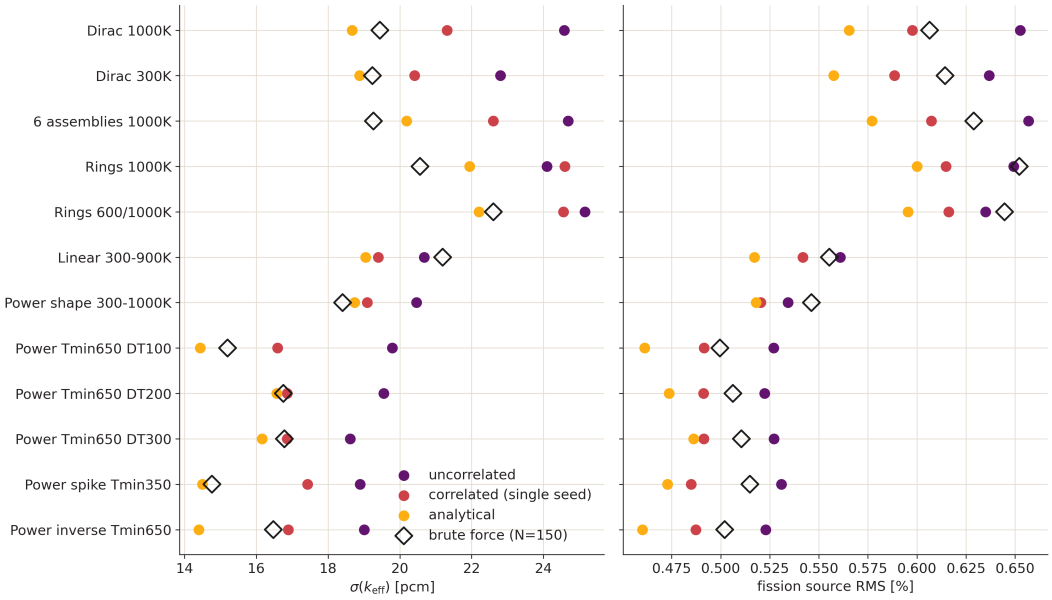

Fig. 7. Database-side uncertainty predicted by the different σ_{interp} quantification methods against the brute-force reference (iv, $N = 150$): k_{eff} (left) and fission-source RMS (right), corrected variant, all twelve cases.

For k_{eff}, the analytical model is the most accurate of the three, reproducing the brute-force uncertainty with a mean absolute deviation of 4.6%, ahead of the single-

seed computed model (8.7%) and well ahead of the uncorrelated model (18.0%). These differences must be read against the sampling uncertainty of the estimates themselves: with 150 replicas the brute-force standard deviation has a relative uncertainty of 5.8%, and the synthetic estimates ($M = 500$) of 3.2%, so even an exact method would show a mean absolute deviation of about 5% (the 95th percentile of this twelve-case mean under noise alone is 7.3%). The analytical model is therefore statistically indistinguishable from brute force on k_{eff}: its mean signed deviation is -2.4%, with a 95% confidence interval of $[-6.3, 2.9]\%$ (bootstrap over the replicas, accounting for the correlation between cases, which share the same databases), and its standard deviation lies within the 95% confidence interval of the brute-force one in 11 of the 12 cases (0.6 exceptions expected by chance). The other two models overestimate the uncertainty significantly, by $+7.3\%$ $[2.6, 13.4]\%$ (single seed) and $+17.6\%$ $[12.7, 24.5]\%$ (uncorrelated). In terms of coverage, 66.8% and 95.3% of the brute-force replicas of k_{eff} over the twelve cases fall within one and two analytical standard deviations of the brute-force mean, against the nominal 68.3% and 95.4%; for the fission source the same test gives 93.6% at two standard deviations, the slight under-coverage expected from the underestimate discussed below. The intra-column correlations are negative, so neglecting them (method i) inflates the propagated k_{eff} variance, while capturing them in closed form (method ii) does not require the additional simulation that measuring them (method iii) does.

On the fission-source RMS the ranking changes: the uncorrelated (3.3%) and single-seed (3.6%) models track brute force more closely than the analytical one (7.3%), which underestimates it in all twelve cases. The cause is not identified. Correlations between coefficients of different columns are weak (Sec. III.E.2), but such pairs are 126 times more numerous than those sharing a column, so a collective effect on local quantities cannot be excluded.

In the following sections only the analytical model and the brute-force method are used. The analytical model is chosen for its agreement with brute force on k_{eff} and for its simplicity; it also gives the strictest acceptance test, since it predicts the lowest σ_{interp} (Fig. 7) and hence the smallest σ_{tot}. For the fission source this window is slightly too narrow, since the analytical σ_{interp} is about 7% below the brute-force value; the zone counts reported below are therefore given under all four models.

IV.B.1. Stability of the Analytical Model

It is reasonable to object that the analytical model is not truly independent of Monte Carlo noise: the closed form of Eq. (12) is evaluated on the mean coefficients a_i of a single database, which are themselves affected by statistical uncertainty. If a different random seed were used, the predicted matrix $\boldsymbol{\rho}_{\text{pred}}$ would differ slightly, and one may ask whether that sensitivity matters.

To test this, $\boldsymbol{\rho}_{\text{pred}}$ at 300 K was rebuilt from each of the 150 independent seeds of the brute-force set and compared, through the relative Frobenius distance of Eq. (16), against the matrix built from the production seed (Table II).

TABLE II
Sensitivity of $\boldsymbol{\rho}_{\text{pred}}$ to the Database Seed (300 K)

Seed-to-seed spread	Relative Frobenius distance
Mean	0.0013
[16, 84] percentile	[0.0013, 0.0013]
Maximum	0.0014

The spread of Table II is more than two orders of magnitude below the distances of Table I; the Monte Carlo noise in the mean FM coefficients is therefore not a meaningful source of uncertainty for the analytical model, and $\boldsymbol{\rho}_{\text{pred}}$ can be treated as effectively deterministic.

IV.C. Uncorrected and Corrected Results Comparison

Figure 8 compares raw and corrected interpolation against the direct reference, before any uncertainty is folded in, and Table III reports the underlying per-case numbers. Averaged over the twelve cases, the CTCR reduces the mean absolute k_{eff} discrepancy from 40.8 to 23.7 pcm (42%) and the mean fission-source RMS discrepancy from 1.27% to 0.94% (26%). Over the ten cases not used to compute the correction ratios, the reductions are from 35.4 to 25.5 pcm (28%) and from 1.33% to 0.94% (29%). The benefit is markedly uneven. It is largest for the sharpest, most localized profiles — Rings 1000 K improves from 57.4 to 1.2 pcm, and 6 assemblies 1000 K from 53.9 to 24.0 pcm — consistent with the CTCR having been calibrated against a sharp central perturbation. For smoother, less localized profiles it is negligible or mildly unfavorable, see e.g. the Power Tmin650 DT200 case worsening

from −41.5 to −52.8 pcm.

TABLE III
Point-Estimate Discrepancy Against the Direct Reference, All Twelve Cases († calibration cases)

	Δk_{eff} [pcm]		Fission-source RMS [%]	
Case	raw	corrected	raw	corrected
Dirac 1000 K†	79.7	13.5	0.972	0.838
Dirac 300 K†	−55.4	−15.4	1.064	0.975
6 assemblies 1000 K	53.9	24.0	1.451	0.794
Rings 1000 K	57.4	1.2	2.411	1.052
Rings 600/1000 K	42.0	19.5	2.084	1.244
Linear 300–900 K	−46.3	−45.8	1.525	1.212
Power shape 300–1000 K	31.7	−7.4	1.828	1.456
Power Tmin650 DT100	11.7	6.0	0.680	0.688
Power Tmin650 DT200	−41.5	−52.8	0.772	0.748
Power Tmin650 DT300	−5.2	−22.2	0.932	0.806
Power spike Tmin350	−18.4	−24.3	0.909	0.775
Power inverse Tmin650	46.0	51.8	0.671	0.670
Mean $\lvert\cdot\rvert$	40.8	23.7	1.275	0.938
Mean $\lvert\cdot\rvert$, excl. †	35.4	25.5	1.326	0.944

IV.D. Results with the Combined Uncertainty

This section presents the same results as the previous one, with a closer look at the uncertainty budget and hence at the acceptability of the results.

Combining σ_{ref} with σ_{interp} from the analytical model in quadrature gives the total uncertainty σ_{tot} of Eq. (5). Under this combined budget, 75% of the corrected cases fall within $|z| < 2$ (median $|z| = 0.94$), against 42% of the raw cases (median $|z| = 2.18$). Over the ten cases not used to compute the correction ratios, the fractions are 70% and 50%, respectively. σ_{interp} is 14–22 pcm against 9–13 pcm for σ_{ref} in every case, so the database term still dominates the budget and both variants are judged against a relatively wide band. The statistical uncertainty of the CTCR calibration, not included in σ_{interp}, was evaluated by propagating the noise of the two Dirac calculations through the correction: 500 realizations of the two Dirac matrices were drawn with the analytical correlation model and the Serpent-reported uncertainties, the ratios $r^{\pm}$ recomputed from each, and the correction reapplied to

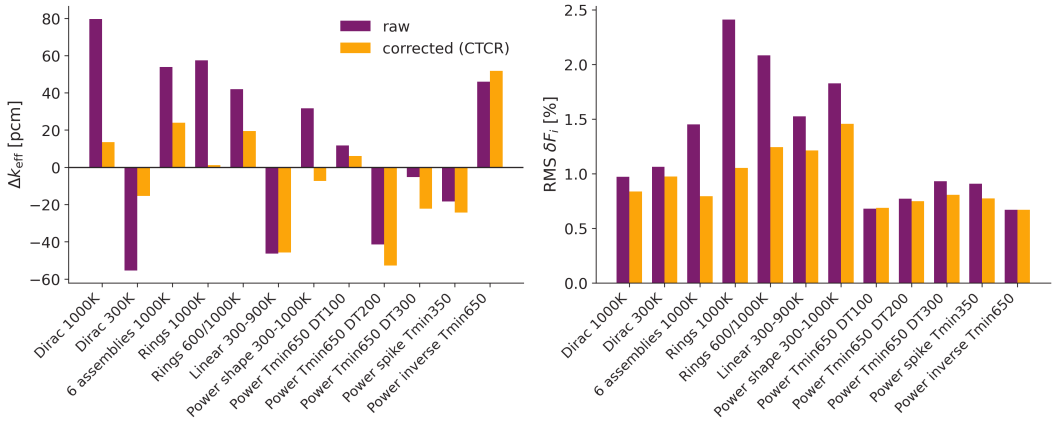

Fig. 8. k_{eff} discrepancy (left) and fission-source RMS discrepancy (right), raw vs. CTCR-corrected interpolation, all twelve cases, with no uncertainty band.

the nominal interpolated matrices of the twelve cases. The resulting uncertainty on the corrected k_{eff} is between 0.1 and 6.2 pcm, at most 28% of σ_{interp}; added in quadrature it increases σ_{tot} by at most 3.3%, and no case changes side of the $|z| = 2$ criterion. Repeating the two Dirac calculations with five times more histories changes the ratios $r^{\pm}$ by about 0.3% and the corrected k_{eff} by at most 5.2 pcm, consistent with this estimate.

Rather than collapsing this budget to a single $\pm 2\sigma_{\text{tot}}$ number per case, Fig. 9 shows, for k_{eff}, all $M = 500$ analytical-model realizations (gray) and all 150 independent brute-force replicas (blue open circles), together with the reference case's own $\pm 2\sigma_{\text{ref}}$ (pink band) and the raw and corrected point estimates (black and red).

Figure 10 reports the numbering of the core assemblies, used as zone index in the fission-source figures. For the fission source (Fig. 11) the same logic is applied zone by zone, but only the $N = 150$ independent brute-force replicas are shown, raw in gray and corrected in shades of red: the $M = 500$ synthetic analytical realizations of every F_i would add clutter without adding information beyond Fig. 7. A corrected point is recolored gold when it falls inside the $\pm 2\sigma_{\text{ref}}$ band *and* the matching raw point, of the same zone, was outside it. Every other corrected point — already consistent with noise in raw, or still not explained by noise even after correction — keeps the base red. This isolates the population the correction is meant to act on, rather than diluting the plot with every in-band point regardless of whether

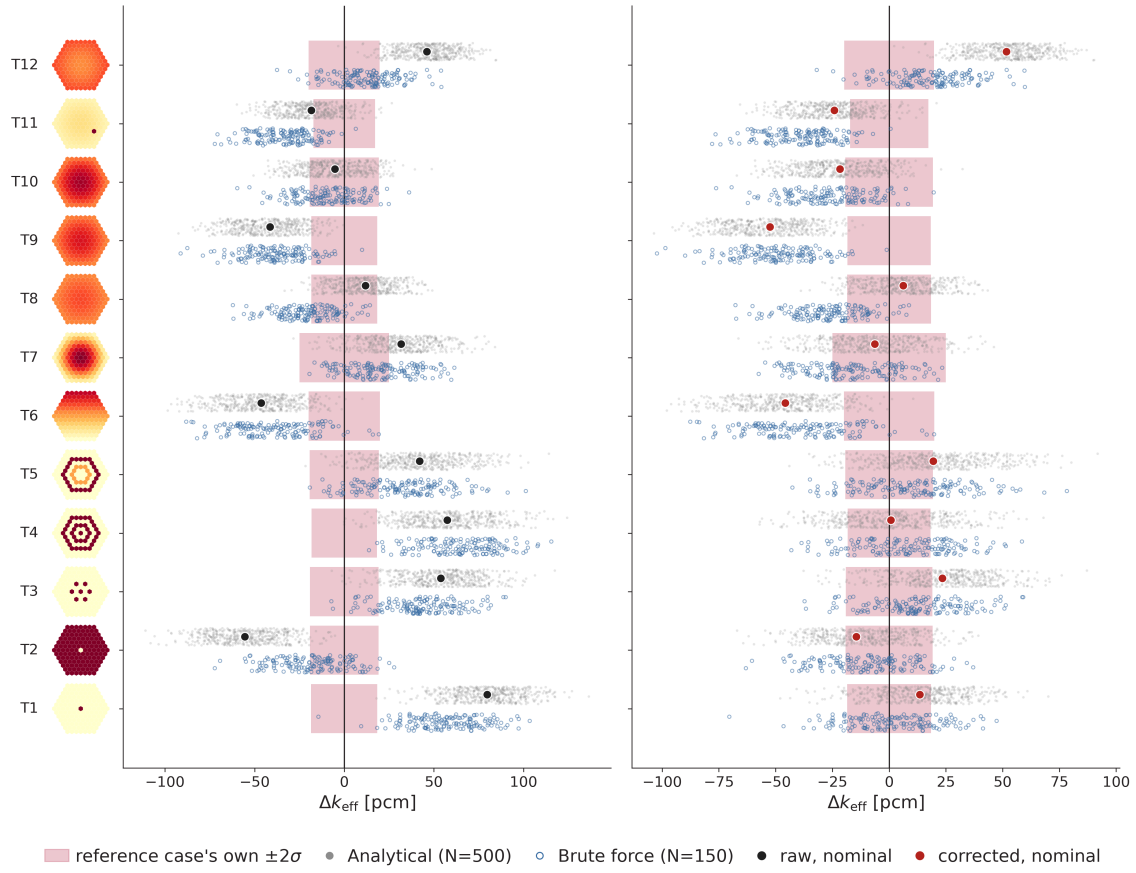

Fig. 9. k_{eff} discrepancy, raw (left) and corrected (right): every analytical-model realization (gray, $M = 500$) and every true independent brute-force replica (blue open circles, $N = 150$), with the reference case's own $\pm 2\sigma_{\text{ref}}$ (pink band) and the point estimate (filled circle), all twelve cases.

raw needed fixing there at all. For brevity, Fig. 11 shows only the first six cases; they include cases where the CTCR strongly affects F_i and cases with smoother temperature differences, where it is less needed. The remaining six cases are reported in Fig. 12 (Appendix).

In several cases of Fig. 11 the whole replica distribution, raw and corrected alike, is offset from zero. This offset is itself informative: it is a bias that no amount of resampling the database can correct. Besides the local-temperature approximation targeted by the CTCR, two other sources of bias were examined. The first is the temperature of the region surrounding the assemblies, which in the database follows the uniform temperature of each reference point, whereas in the test cases it is

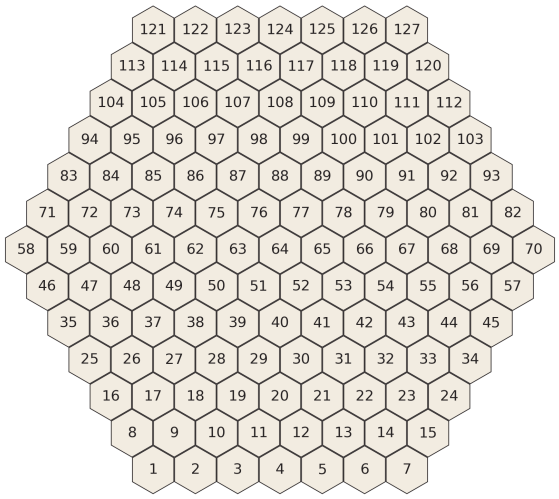

Fig. 10. Hexagon-index map (1–127) referenced by the zone-index axis of Fig. 11 and Fig. 12; hexagon 64 is the central assembly C of the Dirac calibration cases.

fixed at 300 K (at 1000 K in the cold-center Dirac case, whose background is at 1000 K). Given the low density of this region (Sec. II), the difference is expected to have little effect. An alternative database with this region at 300 K at all reference points changes the raw k_{eff} discrepancy by between -45 and $+32$ pcm, with a mean of -0.1 pcm, consistent with the statistical noise of the two databases (the RMS of the change is 0.97 times the value expected from that noise). In a reflected core the reflector temperature would not be negligible, and the database and the reference calculations should then use the same convention for it. The second is the finite temperature grid. The five cases with the sharpest fields (Dirac, six assemblies, rings) contain only database temperatures, so for them the interpolation in temperature is exact and the raw discrepancy comes from the local-temperature approximation and Monte Carlo noise only.

Where the corrected replica cloud is centered close to zero — Dirac 300 K, Linear 300–900 K, and most of the smoother power-shaped profiles — the residual deviation from the reference is statistically indistinguishable from Monte Carlo noise, and no further correction is warranted. Where it is not — the localized spikes in Dirac 1000 K, 6 assemblies 1000 K, and the Rings cases — the deviation exceeds what either noise source can account for, and it therefore indicates an uncorrected bias rather than statistical scatter. Because the database-noise model has itself been

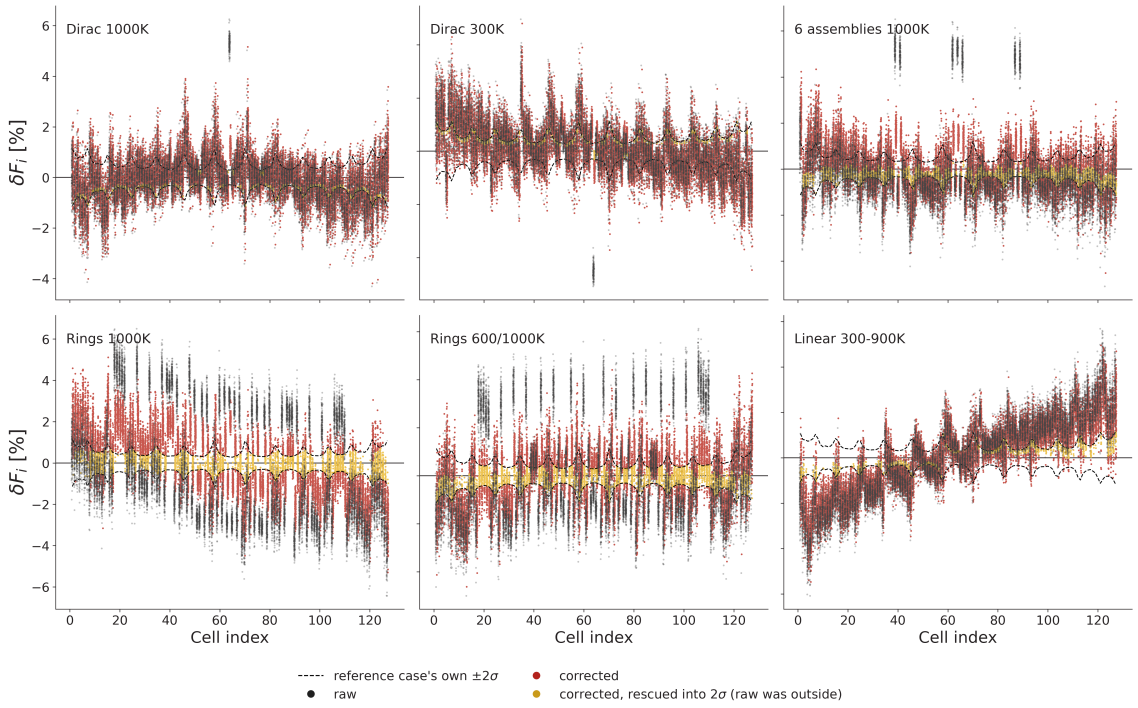

Fig. 11. δF_i of the raw (dark gray) and corrected (red) cases, for each of the 150 brute-force replicas. A corrected point is shown in gold when it falls inside the reference case's $\pm 2\sigma_{\text{ref}}$ band while the raw point of the same zone and replica falls outside it.

validated against brute force, it is not the plausible origin of these residuals; since these cases contain only database temperatures, the residuals are not interpolation errors either: they are the part of the error of the local-temperature approximation that the CTCR does not correct. Which part can be inferred from the reach of the correction. Calibrated on a single perturbed assembly, it resolves the effect of a temperature difference only at short distance: the Dirac perturbation changes the coefficients a_{Cj} that enter Eq. (3) by about 6% up to the second ring, and beyond it by less than their statistical noise. On k_{eff} the correction is therefore effective where an assembly and its first neighbors differ strongly in temperature, as in the sharp cases, and nearly inactive in the smooth fields, where neighboring assemblies differ by a few tens of kelvin (50 K per row in the linear gradient) while the differences across the core reach several hundred kelvin. The part of the error due to the temperature of the more distant assemblies crossed by the neutrons is left uncorrected. This is consistent with the cases where the correction changes k_{eff} little

(linear gradient) or increases its discrepancy (three of the four Tmin650 profiles), while the fission-source discrepancy still decreases or is unchanged there (Table III).

A global metric can be misleading about how the correction performs locally. The CTCR improves both aggregated figures of merit, k_{eff} and the fission-source RMS, but a global metric averages over all 127 zones and cannot distinguish a case that improves a little everywhere from one that improves strongly in the one zone that matters and not at all elsewhere. Table IV gives the zone-by-zone count for each case: for each of the twelve cases, the number of zones (out of 127) whose fission-source point estimate is statistically consistent with noise ($|\delta F_i| < 2\sigma_{\text{tot}}$), raw versus corrected, under each of the four σ_{interp} models.

TABLE IV
Number of Zones (out of 127) Whose Fission-Source Deviation Is Statistically Consistent with Noise ($|\delta F_i| < 2\sigma_{\text{tot}}$), Raw (r) vs. Corrected (c), by Case and σ_{interp} Model

	(i) Uncorr.		(ii) Analyt.		(iii) Meas.		(iv) Brute force	
Case	r	c	r	c	r	c	r	c
Dirac 1000 K	116	118	110	109	112	112	113	115
Dirac 300 K	110	106	103	100	105	100	107	105
6 assemblies 1000 K	103	114	102	103	103	106	103	112
Rings 1000 K	41	84	36	77	37	79	40	88
Rings 600/1000 K	55	87	51	85	55	85	55	88
Linear 300–900 K	66	90	62	82	65	88	66	89
Power shape 300–1000 K	61	77	59	76	60	76	60	76
Power Tmin650 DT100	118	116	115	114	116	114	117	115
Power Tmin650 DT200	117	119	113	113	114	114	115	115
Power Tmin650 DT300	97	104	96	101	93	102	97	103
Power spike Tmin350	109	109	101	106	103	107	107	107
Power inverse Tmin650	120	119	114	115	117	116	117	116

The pattern is consistent with the point-estimate result of Table III, but is sharper here: for the Rings cases the corrected count roughly doubles (e.g., Rings 1000 K rises from 36–41 to 77–88 zones out of 127, consistently across all four models), while for the cases whose raw count is already high (Dirac 300 K, Power Tmin650 DT100/DT200) the shift is small or even mildly unfavorable. The Appendix collects the remaining six cases; among them Power shape 300–1000 K shows one of the largest improvements of the whole set, its corrected count rising from 59 to 76 zones (Fig. 12). Local accuracy matters in practice: in many transient analyses the quantity of interest is not a core-averaged parameter but the most critical

point in the system, such as a localized hot spot, and there the interpolated model is judged by its prediction at that location rather than by k_{eff} or the RMS error.

V. CONCLUSIONS

The interpolated and CTCR-corrected fission matrix inherits statistical noise from the database it is built from, but — unlike a direct MC calculation — its fundamental eigenvalue and eigenfunction have no uncertainty of their own to read off from any single simulation output; it has to be constructed by resampling the database itself. Of the three cheap ways of doing this, a zero-replica analytical model, derived from the branching structure of the fission process under the simplifying assumption that only same-column coefficients are correlated, reproduces the k_{eff} uncertainty of an $N = 150$ brute-force reconstruction at least as well as an approach that requires measuring a full correlation matrix. Both improve markedly on assuming no correlation at all. On the fission-source RMS, the uncorrelated and measured-correlation approaches perform comparably to each other and better than the analytical model, which underestimates it, though all three remain within the same order of magnitude of brute force. Folding both uncertainty sources into a single budget, the CTCR's point-estimate benefit persists once both noise sources are accounted for, and most of the remaining case-by-case discrepancies are statistically consistent with this combined floor. Where they are not, the residual is attributed to the local-temperature approximation, which the CTCR corrects only in part, and, where the field contains intermediate temperatures, possibly to the temperature grid, rather than to the fission-matrix noise model used to characterize it — a distinction the budget makes possible, and a target for future work independent of the covariance itself.

Over all twelve test cases, the mean absolute error of the raw interpolation is 40.8 pcm on k_{eff} and 1.27% on the fission-source RMS; with the CTCR it falls to 23.7 pcm and 0.94%. Over the ten cases not used to compute the correction ratios, the reductions are from 35.4 to 25.5 pcm and from 1.33% to 0.94%.

The analytical model assumes that each history contributes to at most one destination, as with the analog capture and fission used here. Non-unit weights only change the normalization of the variance (Eq. (13)), but variance-reduction techniques such as implicit capture, implicit fission or particle splitting let one history,

or its descendants, contribute to several destinations, and change the covariance structure. Extending the model to these schemes, and to correlations between coefficients of different columns, is left to future work; where a closed form is not available, the cycle-based estimate of method (iii) remains applicable.

ACKNOWLEDGMENTS

The authors would like to acknowledge that the research activity of the PhD project is sponsored by the Italian funding plan "Piano Nazionale Ripresa e Resilienza" (PNRR) and by Westinghouse Electric Company to support the development of methods for advanced reactors. This work has been carried out during the author's internship period at Westinghouse Electric Company. The methodology developed is an extension of the studies previously made during the author's internship activity at Argonne National Laboratory under the supervision and support of Dr. Nicolas Stauff and his research team.

APPENDIX: FISSION-SOURCE DEVIATION, TEST CASES 7–12

Fig. 11 in the main text shows the first six cases of Table III (the two Dirac, the six-assembly, the two ring and the linear-gradient fields). Fig. 12 gives the same plot, in the same style, for the remaining six cases, which form the power-shaped family.

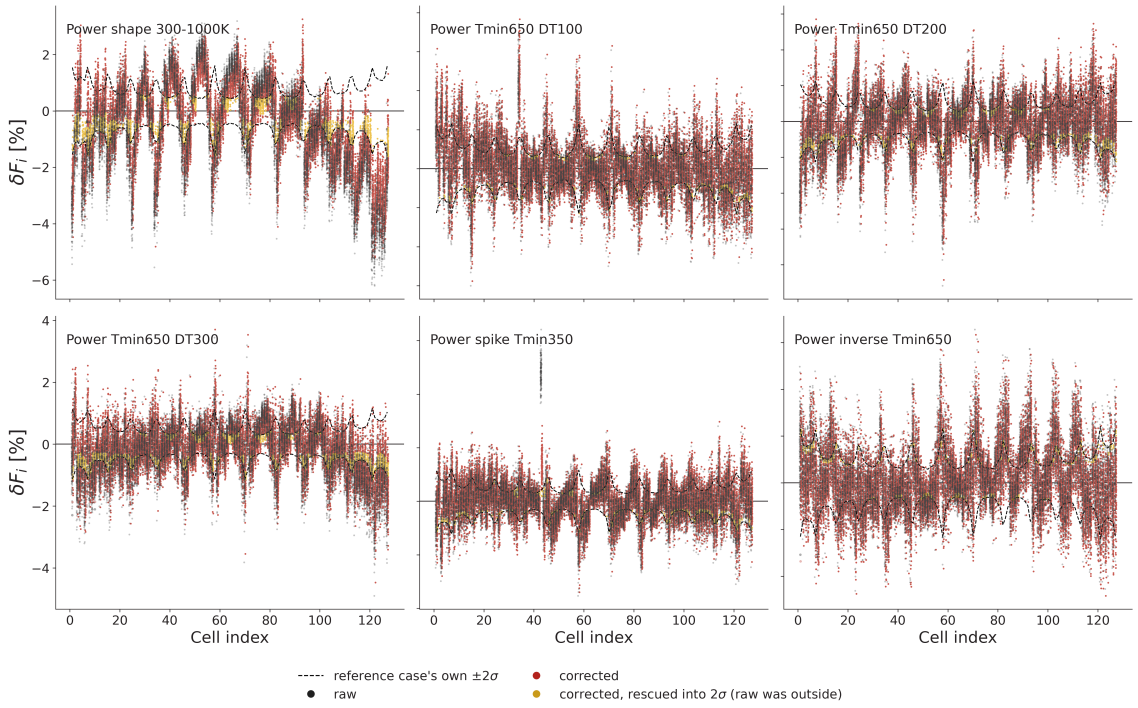

Fig. 12. As Fig. 11, for test cases 7–12.

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